

Lab 3 Worksheet — Mobile Connectivity — Bluetooth, NFC, Hotspot and Cellular Configuration

Name: ____ Date: ____

Exam objective: Configure and troubleshoot mobile connection methods including Bluetooth pairing, NFC, hotspot tethering and cellular data settings (Core 1 objectives 1.3 and 1.4).

Record as you go

Step	What you did	What you observed
1	On your mobile device, open Settings and record the exact menu path to Bluetooth, hotspot, NFC and mobile data settings.	
2	Complete the five-step Bluetooth pairing sequence with a second device: enable Bluetooth, enable pairing mode, discover, enter the PIN or confirm the passkey, then test that data actually flows.	
3	Break the pairing by disabling pairing mode on the target device mid-discovery, record the exact symptom, then re-pair and note which step recovered it.	
4	Enable the mobile hotspot, connect the laptop to it, and record the SSID, the security type in use and the IP address the laptop receives.	
5	Open IP Calculator at https://alfredang.github.io/ipcalculator/ and enter the laptop's hotspot IP address with its netmask in the IPv4 tab.	
6	Record from the calculator output the network address, broadcast address, usable host range and total usable hosts, and confirm the hotspot is using a private RFC 1918 range.	
7	Enable airplane mode, then selectively re-enable	

Step	What you did	What you observed
	Wi-Fi only, and record which connection methods survive and which do not.	
8	Check the cellular data settings and record the APN, the network type currently in use (LTE, 5G) and whether data roaming is enabled.	
9	If your device supports NFC, enable it and record the maximum working distance you observe, then compare it against the 4 cm specification.	
10	Build the troubleshooting flowchart: for each of 'no Bluetooth pairing', 'hotspot connects but no internet', 'NFC not detected' and 'no cellular data', give the checks in the order you would run them.	

Verification

Success criterion: IP Calculator confirms the hotspot address is in a private range with the correct usable host count; your flowchart gives an ordered check sequence for all four symptoms; and each broken-and-fixed cycle is documented with its symptom.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
